e office :
SRM/SRMEC K/25/6161

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To
The Registrar
SRM Institute of Science and Technology
Kattankulathur, Tamil Nadu – 603203

Date: 26th June 2025
E-office ID: **SRM/SRM EC K/25/6161**

Request for Signature on Inception Report for NIDM Research Project by June 27, 2025

Respected Sir,

With reference to the National Institute of Disaster Management (NIDM) communications vide Letter No. NIDM/GIDRR/123/IUIN-DRR/2021, dated February 07, 2025, and February 13, 2025, regarding the Research Project titled **"Development of a GLOF Warning System through Strong Endurance Drone-based Long-Range (LoRa) Communication for the Upper Catchment of the Teesta River"**, we, the undersigned, have accepted all the Terms and Conditions of the Research Project/ Study on Disaster Management: Innovation & Action Research for the Financial Year 2024-25.

The project is being conducted under the aegis of SRM Institute of Science and Technology (SRMIST), Kattankulathur, which is the designated parent institution for receiving the research grant. The project was registered on February 20, 2025, and the **inception report was initially submitted to NIDM on March 31, 2025**. Upon receiving further requests from NIDM for certain changes and clarifications, we have **updated and revised the inception report** accordingly and are now resubmitting it for the release of first instalment.

In line with the requirements of the research grant process, we kindly request your esteemed office to review and provide the necessary verification and endorsement of the revised Inception Report, including the official seal and signature of the Registrar. This will enable its timely submission to NIDM and facilitate the release of the first instalment of the research grant for the Financial Year 2024-25.

Your prompt support in this regard will be greatly appreciated, as it is vital for the official approval and subsequent processing of the research grant by NIDM.

Thank you for your consideration and support.

AR (A)

28/6/25

Yours sincerely,

Dr. Manoj Kumar Rana
Research Assistant Professor
Department of Computing Technologies
CET, SRMIST, Kattankulathur

Dr. Snehasish Ghosh
Assistant Professor
Department of Computing Technologies
CET, SRMIST, Kattankulathur

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INCEPTION REPORT

Project Title: Development of a GLOF Warning System through Strong Endurance Drone- based Long-Range (LoRa) Communication for the Upper Catchment of the Tista River

1. Executive Summary:

The Himalayan Mountain region is facing a Glacial Lake Outburst due to reducing permafrost, resulting in the formation of new lakes with the circumferences of unstable natural dams, which can be easily breached by rainfall, seismic activity, landslides, or avalanches. Different satellite images such as Sentinel-2(10m), Resourcesat- 2 (LISS-IV) data (5 m) can be used for glacial lake morphological study at preliminary level. Due to lack of high spatial resolution satellite image, unfavourable atmospheric conditions, insufficient high-resolution data to achieve precise digital terrain model and deficiency of authenticated geospatial data for AI/ML training, these GLOF studies might not be entirely sufficient. Combining these images with real-time imagery offers a strong approach for detailed and comprehensive glacial lake morphological studies. Real-time images through drone based remote sensing processes have grown for practical solutions. However, the Himalayan Mountain region encompasses many physical barriers such as rugged terrain, unavoidable wind speed, and the absence of infrastructure that makes the area inaccessible for ground based study. Our objective is threefold. First, development of a fixed-wing, inexpensive, and high-endurance drone equipped with Long-range (LoRa) communication technology (open-source) kind of communication technology and post-processed kinematics from GPS for investigating ice fracturing and glacial lake outbursts. Second, developing a machine learning and AI technique to integrate and analyze data from both sources. Third, implementation of a 4G/5G-based warning system, broadcasting any emergency or important information based on the decision made by our analysis.

2. Introduction:

The project is illustrated below with a brief description and the problem statement.

i. Brief of project:

The Himalayan region is increasingly at risk of Glacial Lake Outburst Floods (GLOFs) due to melting permafrost and the formation of unstable glacial lakes. Current satellite imagery lacks the resolution and real-time capabilities needed for precise monitoring, while challenging terrain limits ground-based studies.

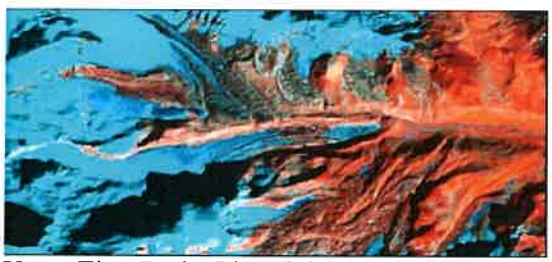
To address these challenges, this project proposes a threefold solution:

a) **Fixed-Wing Drone Development** – An inexpensive, high-endurance drone equipped with Long-Range (LoRa) communication type communication module and GPS post-processed kinematics for real-time monitoring of ice fracturing and glacial lake changes.
b) **AI/ML Integration** – A machine learning framework to analyze and integrate data from satellite images and drone observations for enhanced GLOF susceptibility mapping.
c) **4G/5G-based Early Warning System** – A real-time alert system that broadcasts emergency warnings based on AI-driven risk assessments.

By combining advanced drone technology, AI analytics, and robust communication systems, this project aims to improve GLOF monitoring and disaster response, enhancing safety and resilience in the Himalayan region.

ii. Problem statement:

The Himalayan Mountain region is facing a Glacial Lake Outburst due to the reduction of permafrost region in Greater Himalayan region, resulting in the formation of new lakes and its extension with the circumferences of unstable natural dams, which can be easily breached by rainfall, seismic activity, landslides, or avalanches. This is the preliminary condition of uncertain mountain flood and it is the major concern at the present climatic scenario where global warning and climate change increases the melting rate of glacier in higher altitude region. The glacial Lake scenario in different times is shown in Fig. 1. It poses a real threat of downstream flash flood in a river basin where possible impact will be on destruction of settlement and human lives without any alert. In reality, there will not be any control over the occurrence of such natural incident. Even, it would not be possible to prevent its damages unless the relevant factors which cause such disaster are monitored in regular interval. Early warning system (EWS) is one of the key precursor resilience for the GLOF disaster risk reduction. With the advancement of technology, the development of GLOF-EWS overcomes the challenges of natural barriers, atmospheric circumstances, cost efficiency and also reduces the human effort on ground with the inclusion of AI/ML technology. The problems identified and the corresponding objectives set up are presented in Table I.



Upper Tista Basin, Lhonak lake [Dec 1990]



Upper Tista Basin, Lhonak lake [Dec 2023]

Fig. 1 Glacial Lake scenario

Table I: Problems and the corresponding objectives

Glacier Lake Outburst Flood (GLOF)	
Problem	**Objective**
1. Limitations of existing imagery and barrier of Himalayan region - Low spatial resolution (6-10 m per pixel) - Bad weather condition - Rugged terrain, unavoidable wind speed - Ground based system not suitable - Private satellite images are very costly (10-30$/$Km^2$)	**1. Development of a fixed wing, high endurance drone or UAV** **2. Investigating and communicating ice fracturing and glacier lake outburst**
2. Conventional mathematical model (e.g. GIS/DEM) - Static based on elevation and slope - Glacier level change, movement not detected - Limited pattern recognition	**3. Developing an ML/AI technique to integrate and analyze from both satellite and drone images**
3. Data centric non-comprehensive Early Warning System (EWS) - Monitoring same location in same direction - Long time to predict - Lack of relevance in state and district level	**4. Implementation of a 4G/5G based warning system**

3. Project team:
The project team is structured as follows:

i. Details of PI & Co-PI:

- Principal Investigator (PI):

Dr. Manoj Kumar Rana
Research Assistant Professor
Department of Computing Technologies
School of Computing
College of Engineering and technology
SRM Institute of Science and Technology,
Kattankulathur, Tamil Nadu-603203

- Co-Principal Investigator (Co-PI) from SRMIST

Dr. Snehasish Ghosh
Assistant Professor
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School of Computing
College of Engineering and technology
SRM Institute of Science and Technology,
Kattankulathur, Tamil Nadu-603203

- Co- Principal Investigator (Co-PI) from NIDM

Shri Amarjeet Kumar
Assistant Professor
National Institute of Disaster Management (NIDM)
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Plot no. 15, pocket-3, block-b, sector-29,
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ii. Organization chart of project team/partners:

The organization chart of project is shown in Fig. 2.

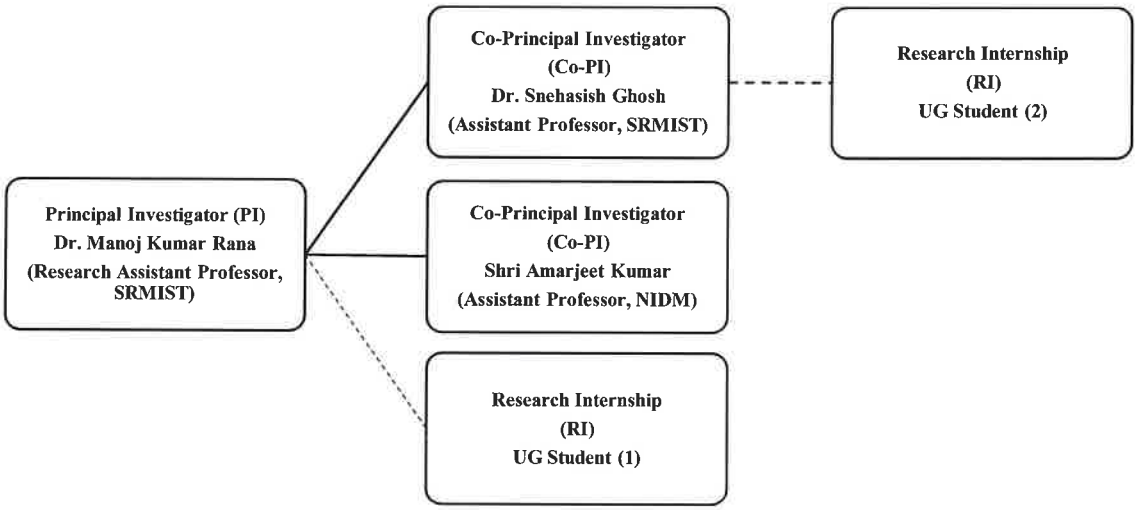


Fig. 2 Organization chart of project team/partners

4. Detailed methodology:

This methodology is outlined through a flowchart shown in Fig. 3, with the details provided step by step below.

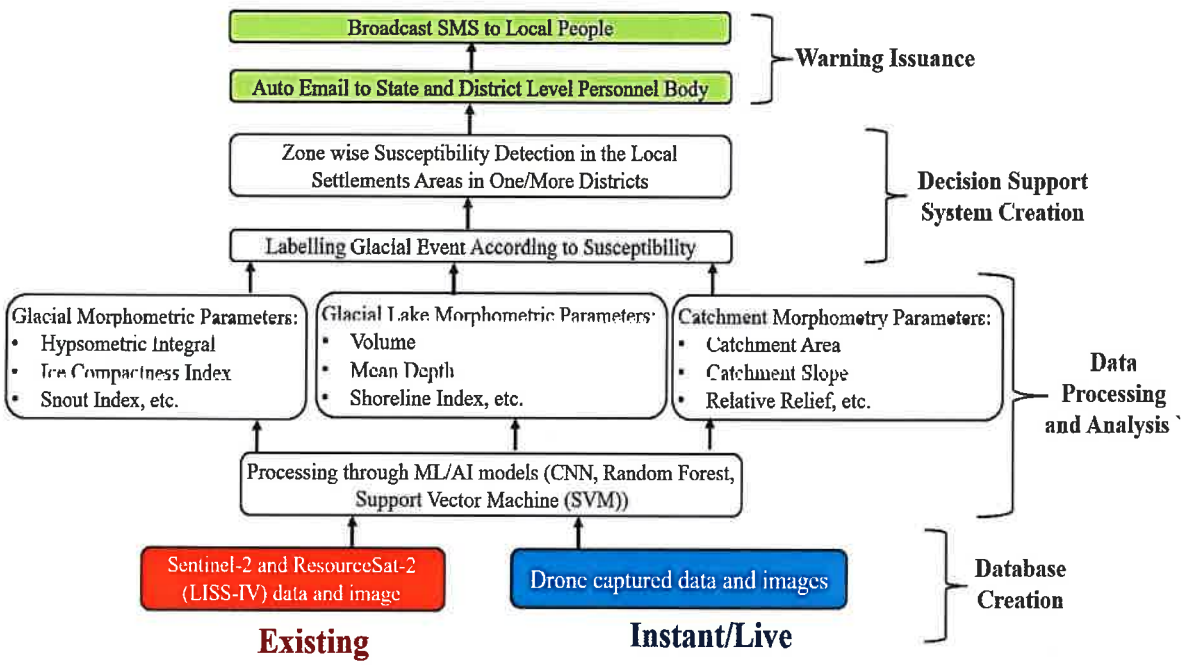


Fig. 3 Outline of the methodology used for the project execution

- **Step 1: Data acquisition**

The data acquisition phase involves collecting images of the target lake(s) from both existing sources and a drone developed by us. However, initially, we will rely on existing sources for post-processing and analysis, as the drone's construction may take some time to complete.

- **Existing sources:** The process starts with data collection from two primary sources: Sentinel-2 (USGS) and ResourceSat-2 (LISS-IV) satellite imagery (NRSC Bhunidhi), which provide pre-recorded data.

- **Drone equipped with camera and communication technology:** Fixed-wing drones excel in Himalayan environments, offering reliable data collection and withstanding Kabat winds of up to 5–10 m/s. They cover up to 50 km without frequent redeployment, while long-range wireless communication enables real-time data transmission in remote areas. GPS post-processed kinematics provide precise geo-referenced ortho-images for accurate GLOF detection. Open-source drone development enhances accessibility, allowing resource-constrained organizations to deploy and customize drones for risk monitoring. By integrating satellite data with high-resolution imagery, drones overcome environmental and temporal constraints for comprehensive GLOF detection and monitoring.
The trajectory of the drone from the base camp to the target lake(s) and back is illustrated in Fig. 4. While our methodology prioritizes in-flight image transmission, images will initially be stored on the camera's SSD card and processed after the drone lands. The in-flight transmission mode will be enabled after field testing, depending on the availability of 4G/5G infrastructure in the area. The overall configuration of the drone and communication modules is shown in Fig. 5.

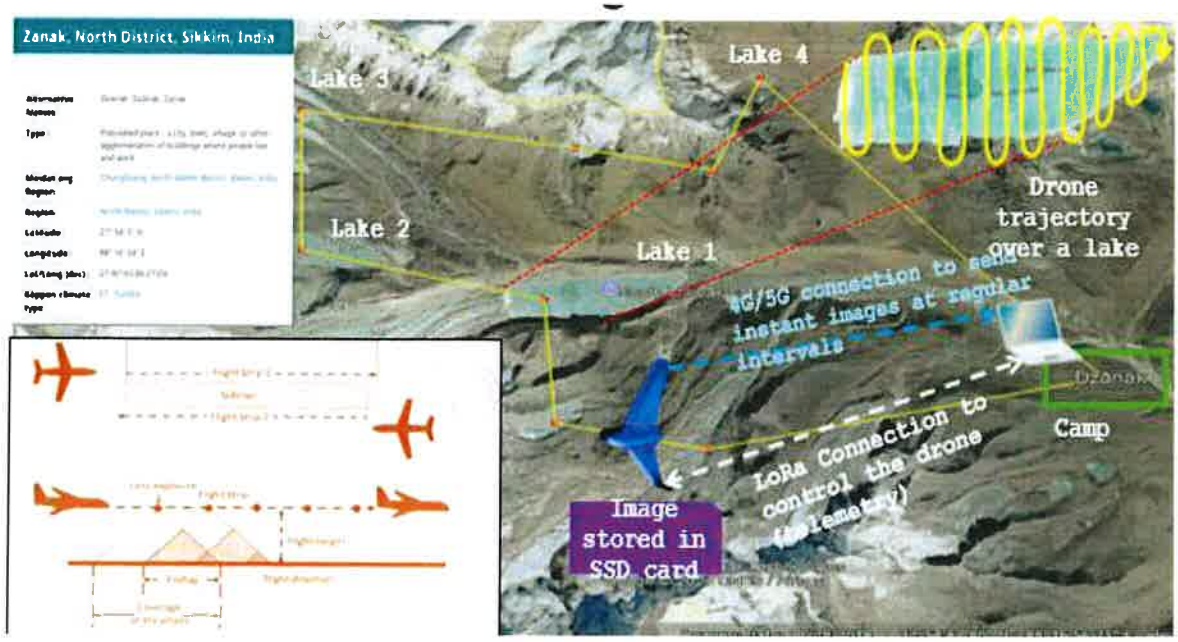


Fig. 4 Drone trajectory from the base camp to the lake(s) and back

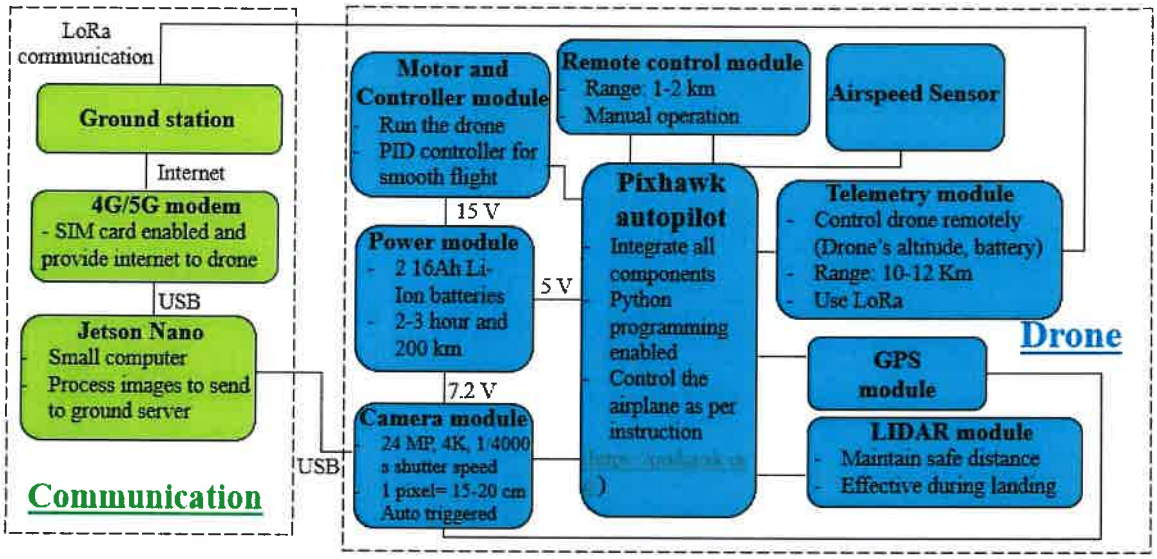


Fig. 5 Configuration of drone and communication modules

- **Step-2: Data Processing and Quantitative analysis**

An ML/AI-based integrated approach is used to analyze data from satellite and drone sources. Advanced models, including Convolutional Neural Networks (CNN), Random Forest, and Support Vector Machines (SVM), detect glacial lakes and monitor their dynamic changes. This solution provides a real-time view of glacial conditions by evaluating historical and real-time data to identify early warning signs of GLOF events.

ML techniques like Random Forest and SVM predict GLOF events, while CNN analyzes drone images to detect anomalies for faster response. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks analyze time-series data to estimate GLOF likelihood. AI/ML automates large-scale monitoring, with unsupervised learning and clustering aiding in identifying new lakes.

This process computes key glacial morphometric parameters such as the Hypsometric Integral, Ice Compactness Index, and Snout Index, along with glacial lake parameters like volume, mean depth, and shoreline index.

- **Step-3: Susceptibility mapping**

The system labels glacial events based on their susceptibility and conducts zone-wise susceptibility detection in affected districts. This information is then used to aid in risk assessment and response planning. To integrate satellite imagery (Sentinel-2, Resourcesat-2) with real-time drone-based remote sensing using AI/ML techniques for glacial lake morphological analysis and early warning. This integration will be guided by three categories of parameters—summarized in Tables II to IV—representing lake, glacier, and catchment characteristics. These parameters will first support susceptibility mapping where threshold for EWS will be generated. Additional spatial and temporal change-detection metrics will help identify potential outburst events and improve model accuracy. Together, these parameters will form the core of an AI/ML framework that combines satellite and drone data for both long-term risk assessment and timely early warning.

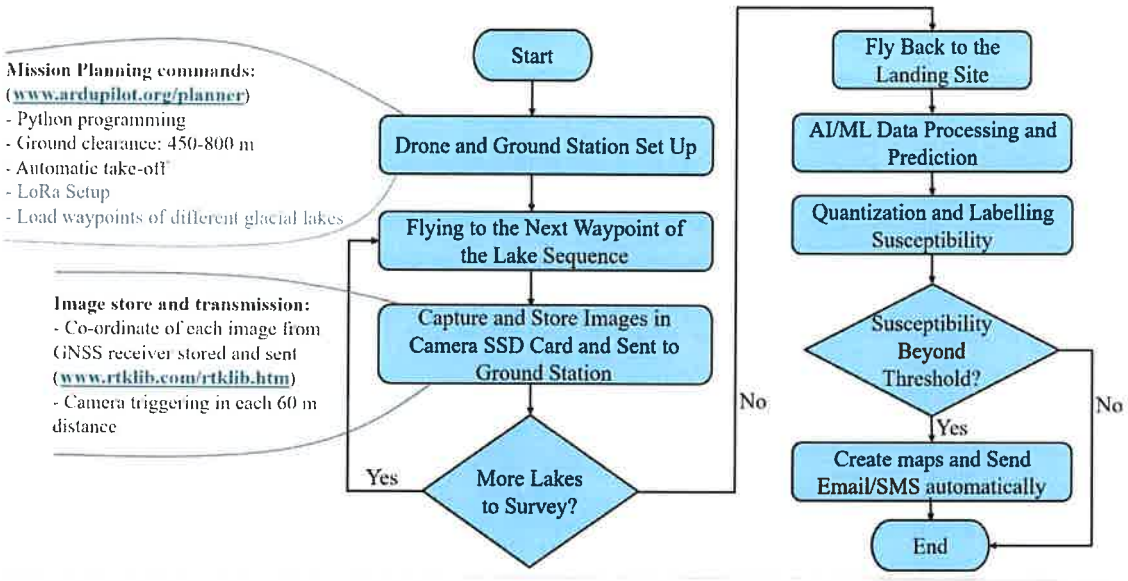


Fig. 6 Procedure for Early warning system

- **Step-4: Development of Early Warning System (EWS)**

The overall procedure of Early warning system (EWS) is shown in Fig. 6. If a potential GLOF is detected, the system triggers an automatic warning issuance process. SMS alerts are sent to local communities, and emails are dispatched to state and district-level authoritics, cnsuring a timely and effective response. By integrating satellite and drone-based remote sensing with AI-driven analysis, this system enhances GLOF monitoring, risk assessment, and disaster preparedness in the Himalayan region.

5. Role & responsibility of project team (PI& Co-PI):

S. No.	Name of the Investigators	Role	Responsibilities in the proposed project
1.	Dr. Manoj Kumar Rana, SRMIST	PI	• To develop an inexpensive, high-endurance fixed-wing drone equipped with Long-range (LoRa) wireless communication and post-processed kinematics (PPK) GPS for monitoring ice fracturing and glacial lake outburst floods (GLOFs). • To implement a 4G/5G-based early warning system that provides real-time alerts on potential GLOF events based on AI-driven analysis. • Mentor One internship student. • Monitoring the progress of the entire work, timely conduct of meeting, administrative tasks, and financial responsibilities. • Helping in AI/ML-Based modelling and analysis
2.	Dr. Snehasish Ghosh, SRMIST	Co-PI	To integrate satellite imagery (Sentinel-2, Resourcesat-2) with real-time drone-based remote sensing using

			AI/ML techniques for enhanced glacial lake morphological analysis. **1. Data Acquisition and processing** • **Satellite Imagery Collection**: Obtain high-resolution multispectral images from Sentinel-2 and Resourcesat-2. • **Drone-Based Remote Sensing**: Deploy UAVs equipped with high-resolution cameras, LiDAR, and thermal sensors to capture real-time data. • **Georeferencing & Calibration**: Align satellite and drone images using GPS and ground control points (GCPs) for accuracy. **2. AI/ML-Based Analysis** • **Feature Extraction**: Identify key morphological features (lake boundaries, water levels, ice cover, debris flow). • **Deep Learning for Classification**: Use CNNs or other deep learning models to classify water bodies, ice, and sediment. • **Change Detection**: Apply temporal analysis using AI to detect changes in lake morphology over time. 3. **Quantification of Morphometric parameters** • **Quantification:** Quantification of Glacial Morphometric and Glacial Lake Morphometric Parameters • **Geospatial database generation:** Geospatial data base generation of morphometric parameters 4. **Mentor One internship student**
3.	Shri Amarjeet Kumar, NIDM	Co-PI	• Coordinate with NIDM and other government agencies for seamless collaboration. • Obtain necessary permissions for fieldwork and secure government support. • Oversee resource allocation and ensure timely execution of tasks. • Supervise project execution, ensuring adherence to timelines and budgets. • Facilitate communication between stakeholders, including government bodies, telecom providers, and research institutions. • Oversee risk assessment and implement mitigation strategies.

6. Objectives and principal research question:

a) Objectives

i. To develop an inexpensive, high-endurance fixed-wing drone equipped with Long-range (LoRa) communication and post-processed kinematics (PPK) GPS for monitoring ice fracturing and glacial lake outburst floods (GLOFs).

ii. To develop a **GLOF susceptibility mapping framework** by integrating quantified **lake and glacial morphometric parameters with catchment-level physiographic variables** using suitable machine learning algorithms.

iii. To establish **region-specific susceptibility thresholds for GLOF-prone zones**, enabling the formulation of an Early Warning System (EWS) for timely risk mitigation and decision-making.

iv. To implement a 4G/5G-based early warning system that provides real-time alerts on potential GLOF events based on AI-driven analysis.

Table II: Glacial Morphometric Parameters

Sl no	Parameter	Formula (Equation Form)	Typical Range	Significance to GLOF Risk	Recent Reference (Post-2020)
1	Hypsometric Integral (HI)	HI = ($\bar{h}$ - h_min) / (h_max - h_min)	0.3 – 0.6	Lower HI → mature basin, more lake formation → higher GLOF risk.	Allen et al., 2022
2	Ice Compactness Index (ICI)	ICI = P / (2√(πA))	>1 (irregular), =1 (circular)	Higher ICI → irregular shape, unstable margins → higher GLOF risk.	Wang et al., 2021
3	Accumulation Ratio (AR)	AR = A_acc / A_total	0.4 – 0.8	Lower AR → retreating glacier → higher GLOF potential.	Bhambri et al., 2021
4	Snout Index (SI)	SI = W_snout / L	0.1 – 0.5	Lower SI → narrow snout, weak moraine → higher GLOF risk.	Nie et al., 2020
5	Elongation Quotient (EQ)	EQ = L / √A	1 – 3	Higher EQ → direct water funneling → increased GLOF risk.	Zhang et al., 2023
6	Relief Ratio (RR)	RR = (h_max - h_min) / L	0.05 – 0.5	Higher RR → steep terrain → more energetic GLOF outbursts.	Rounce et al., 2021
7	Relative Upslope Area (RUA)	RUA = A_upstream / A_total_basin	0.2 – 0.8	Higher RUA → more water input → higher GLOF risk.	Veh et al., 2022.
8	Moraine Dam Slope (MDS)	MDS = Slope angle of moraine dam (degrees)	Variable	Steeper MDS → more prone to failure → higher GLOF risk.	Bazai et al., 2024
9	Lake Expansion Rate (LER)	LER = ΔA / Δt	Variable	Faster LER → unstable lake conditions → higher GLOF risk	Khadka et al., 2021
10	Distance to Glacier Terminus (DGT)	DGT = Distance from lake to glacier terminus	Variable	Short DGT → more ice calving → higher GLOF risk.	Gurung et al., 2021
11	Dam Freeboard (DF)	DF = Elevation_dam_crest - Elevation_lake_surface	Variable	Lower DF → easier overtopping → higher GLOF risk.	Dubey et al., 2023
12	Catchment Area (CA)	CA = Total contributing drainage area (GIS)	Variable	Larger CA → more runoff → increased lake volume → higher GLOF risk.	Zhang et al., 2024
13	Slope of Ice Tongue (SIT)	SIT = Ice tongue slope (degrees)	Variable	Steeper SIT → more calving → higher GLOF trigger potential.	Gurung et al., 2021
14	Glacial Area (GA)	Area of the glacier feeding the lake.	0.1–100 km²	A larger glacier (accumulation + ablation zones) can release larger avalanches	Hu et al., 2022

Table III: Glacial Lake Morphometric Parameters

Sl No	Parameter	Formula / Calculation	Typical Range / Units	Significance to GLOF Risk	Recent Reference (Post-2020)
1	Surface Area (A)	A = measured lake surface area	0.01 – 5 km²	Higher area → more water stored → higher GLOF volume	Nie et al., 2022
2	Surface Volume (V)	V = A × Mean Depth	0.01 – 1.5 km³	Higher volume → greater potential outburst discharge	Zhang et al., 2023
3	Mean Depth	$\bar{D}$ = V / A	1 – 50 m	Greater depth → more energy retained → stronger GLOF	Zhang et al., 2023
4	Maximum Depth	Measured max depth	2 – 100 m	Deeper lakes → higher dam pressure → higher failure risk	Wang et al., 2021
5	Shoreline Length (L)	Measured lake perimeter	0.5 – 15 km	Longer shoreline → more irregularity → dam instability	Li et al., 2023

6	Shoreline Development Index	SDI = L / (2√(πA))	1 – 3	Higher SDI → complex shape → unstable dam margins	Singh et al., 2022
7	Freeboard Height	Vertical height from water surface to dam crest	0.5 – 10 m	Lower freeboard → easier overtopping → higher GLOF risk	Chen et al., 2024
8	Lake Elevation	Measured altitude above sea level	3000 – 5500 m	Higher elevation → rapid melt triggers → more GLOFs	Müller et al., 2023
9	Lake Length	Maximum horizontal length	0.1 – 5 km	Longer lakes → channeled flow → directed flood surge	Xu et al., 2023
10	Lake Width	Maximum horizontal width	0.05 – 2 km	Wider lakes → more volume → broader flood impact	Kumar et al., 2022
11	Length-to-Width Ratio (L/W)	Lake Length / Lake Width	1 – 5	Higher ratio → steep terrain → more energetic GLOFs	Patel et al., 2023
12	Catchment Area	Measured area draining into lake	1 – 100 km²	Larger area → more melt input → bigger GLOFs	Zhao et al., 2023
13	Ice Contact Area	Glacier-lake contact zone	Measured via remote sensing	Larger contact → higher avalanche risk → GLOF trigger	Singh et al., 2024
14	Sediment Influx / Turbidity	Measured sediment load or turbidity (NTU)	10 – 1000 NTU	High turbidity → dam weakening → easier failure	Wang et al., 2022
15	Dam Type & Composition	Moraine / ice / bedrock / mixed	Qualitative	Moraine dams → least stable → highest failure risk	Bajracharya et al., 2023
16	Dam Width (crest)	Distance across the top of the dam (measured in meters)	20–30 m	Width of the dam at crest (measured from imagery). Narrow dams are more failure-prone	Rounce et al., 2021

Table IV: Catchment Morphometry Parameters for GLOF Susceptibility (EWS parameters and its typical threshold value at catchment level warning)

Parameter	Formula / Description	Typical Threshold Range	GLOF Risk Interpretation	Reference
Catchment Area (A)	Total drainage area (km²)	5 – 200 km²	Larger area may indicate greater water volume potential	Worni et al. (2013), ICIMOD Reports
Catchment Slope (S)	S = (Hmax - Hmin) / L (in %)	10% – 40%	Steeper slopes lead to faster water and debris movement	Rounce et al. (2016)
Relief (R)	R = Hmax - Hmin	300 – 1500 m	Higher relief increases potential energy and risk	Veh et al. (2018)
Relative Relief (RR)	RR = R / sqrt(A)	5 – 15 m/km	High relative relief = greater instability	ICIMOD Guidelines
Drainage Density (Dd)	Dd = Ld / A	0.5 – 3.0 km/km²	Higher Dd → efficient runoff → higher peak flow	Shrestha et al. (2010)
Melton Ruggedness Number (MRN)	MRN = (Hmax - Hmin) / sqrt(A)	0.3 – 1.5	High MRN → higher energy floodwaters, debris flow risk	Huggel et al. (2004), Worni et al.
Lake to Glacier Distance	Horizontal distance from lake to glacier snout (km)	< 1 km = high risk	Shorter distance increases chance of ice/debris input	Emmer and Vilímek (2013)
Aspect (Direction)	Direction of lake/glacier facing (N, S, etc.)	South / SE-facing – higher melt	Affects melt rate, influencing lake volume	Racoviteanu et al. (2011)
Proximity to Settlements	Distance (km) from lake to nearest human settlement	< 10 km = high susceptibility	Closer proximity = greater human risk	Worni et al. (2013)
Form Factor (Ff)	Ff = A / L²	0.1 – 0.5	Lower Ff → elongated basins → faster response times	Strahler (1957)
Surface Temperature Index (STI)	Normalized surface temperature: STI = (Ts - Tmin) / (Tmax - Tmin) (Ts: pixel temperature)	0.6 – 1.0 (normalized) or > 5°C (raw)	High STI → increased glacial melt → lake volume rise → moraine destabilization	Wang et al., 2012; Raj et al., 2013

To support automated GLOF detection using AI/ML, a set of basic spatial and temporal rules is used during data preprocessing and labeling. These include a statistical threshold $T(C) = 0.5^{2k}$, where k is the number of satellite images before and after a detected change, helping to confirm that the change is real and consistent. A minimum connected area (e.g., 4 or more pixels) is also applied to filter out random noise. While Tables II, III, and IV define static parameters for lake, glacier, and catchment characteristics used in susceptibility modeling, these dynamic metrics help track ground level changes over time. Together, they improve training data quality and support reliable early warning when combined with multi-temporal imagery from sources like Sentinel-2 or Resourcesat-2.

Principal research question

"How can a fixed-wing, low-cost, high-endurance drone equipped with LoRa communication, GPS post-processed kinematics, and AI-driven data analysis enhance the monitoring, prediction, and early warning of ice fracturing and glacial lake outburst events?"

7. Scope of project:

The project encompasses the development and implementation of an integrated monitoring and early warning system for Glacial Lake Outburst Floods (GLOFs) in the Himalayan region. It involves the use of satellite and drone-based remote sensing technologies to enhance data collection and risk assessment. The scope includes designing a fixed-wing, high-endurance drone equipped with long-range communication and post-processed GPS kinematics for real-time glacial lake monitoring.

Additionally, the project covers the integration of machine learning and AI techniques to analyze satellite and drone imagery, enabling precise detection of glacial lake changes and potential GLOF risks. The deployment of a 4G/5G-based early warning system is also within the scope, ensuring the timely dissemination of alerts. The approach is designed to be scalable, cost-effective, and adaptable, supporting long-term glacial hazard mitigation efforts in remote and vulnerable regions.

8. Stakeholders:

A multi-level stakeholder approach is essential for the successful implementation of this Glacial Lake Outburst Flood (GLOF) monitoring and early warning system. The key stakeholders include:

1. National-Level Stakeholders

- **National Institute of Disaster Management (NIDM)** – Responsible for research, training, and policy development for disaster risk reduction.
- **National Disaster Management Authority (NDMA)** – Oversees disaster preparedness and response at the national level and ensures the implementation of early warning systems.
- **Ministry of Environment, Forest, and Climate Change (MoEF)** – Supports research and policy initiatives for climate change adaptation and environmental conservation.
- **Indian Space Research Organisation (ISRO)** – Provides satellite-based remote sensing data for glacial lake monitoring.

2. State-Level Stakeholders

- **State Disaster Management Authorities (SDMAs)** – Implement disaster mitigation and early warning strategies at the state level.
- **State Remote Sensing Agencies** – Assist in geospatial data analysis for GLOF monitoring.
- **State Departments of Water Resources, Environment, and Climate Change** – Support research, policy, and implementation of climate adaptation strategies.

3. District-Level Stakeholders

- **District Disaster Management Authorities (DDMAs)** – Coordinate local disaster preparedness and response efforts.
- **District Administration (District Collectors & Local Authorities)** – Oversee emergency response and communication with communities.
- **Local Law Enforcement & Emergency Services** – Assist in evacuation and disaster response operations.

4. Scientific & Research Institutions

- Universities, climate research centers, and remote sensing labs working on glacial monitoring, AI/ML modeling, and disaster risk reduction.

5. Local Communities & NGOs

- **Community-Based Organizations (CBOs) & NGOs** – Conduct awareness campaigns, assist in emergency response, and support community resilience.
- **At-Risk Populations in Himalayan Regions** – Direct beneficiaries of the early warning system, requiring training and preparedness strategies.

6. Technology & Infrastructure Partners

- **AI & Drone Technology Developers** – Work on drone-based remote sensing and predictive analytics.
- **Telecommunication Providers (4G/5G Networks)** – Ensure connectivity for real-time data transmission and emergency alerts.

By integrating national, state, and district-level authorities with scientific institutions, local communities, and technology providers, this project aims to create a comprehensive and effective GLOF monitoring and early warning system for the Himalayan region.

9. Timeline and delivery schedule:

ACTIVITIES	I YEAR				II YEAR			
	3	6	9	12	3	6	9	12
1. Student selection (UG level), Equipment purchase, Satellite data acquisition								
2. Field visit and data collection: Drone image collection, Database creation, Ground based evidence								
3. Image processing through software: Object detection and image classification								
4. Glacial and glacial lake Morphometric analysis and quantification								
4. ML based model and decision Support System								
5. Early Warning System for GLOF: Monitoring of Dynamic changes of glacial lake, Checking warning at various level, Message broadcasting using 4G/5G network								
6. Report Submission								
7. Research publications and patent work								

10. Plan of Action:

- **Objective 1: Development of a Fixed-Wing, High-Endurance Drone**

i. **Design and Prototyping**
   - Develop a fixed-wing drone optimized for high endurance and long-range operations.
   - Utilize lightweight composite materials for enhanced flight time and durability.
   - Integrate solar-assisted power solutions to extend operational duration.

ii. **Technology Integration**
   - Equip the drone with Long-range (LoRa) communication technology for efficient data transmission.
   - Implement Post-Processed Kinematics (PPK) GPS for high-precision geolocation.
   - Include multispectral and LiDAR sensors for improved glacial monitoring.

iii. **Testing and Optimization**
   - Conduct flight tests in controlled environments to evaluate endurance, stability, and data acquisition.
   - Optimize drone navigation algorithms to counteract rugged terrain and high wind speeds.
   - Ensure real-time data transmission capabilities for timely decision-making.

- **Objective 2&3: Development of AI/ML-Based Analytical Techniques**

i. **Data Collection and Processing**
   - Gather satellite images from Sentinel-2 and Resourcesat-2 (LISS-IV) for preliminary analysis.
   - Acquire real-time imagery from drone-based remote sensing for high-resolution data.
   - Process and clean geospatial datasets to ensure quality inputs for AI/ML models.

ii. **Machine Learning and AI Model Development**
   - Train AI/ML models to detect ice fracturing, glacial lake formation, and potential outbursts.
   - Use deep learning techniques for precise segmentation of glacial lakes and unstable natural dams.
   - Develop predictive models integrating various geospatial and climatic parameters.
   - Reference parameter categories listed in Tables II–IV to guide model feature selection, covering both susceptibility indicators and EWS thresholds derived from satellite and drone-based inputs.
   - **Proposed Framework for GLOF Susceptibility Mapping and Early Warning System (EWS): 1) Quantification of Lake Morphological Parameters:** Table III parameters will be quantified using remote sensing and GIS techniques. **2) Quantification of Glacial Morphometric Parameters:** Table II parameters will be analyzed and quantified. **3) Integration Using Machine Learning Algorithms:** The quantified lake and glacial morphometric parameters will be integrated using appropriate machine learning algorithms to generate a GLOF susceptibility mapping at the catchment level. This integration will also include **catchment physiographic parameters** such as slope, aspect, relative relief, proximity to glaciers/lakes, and other terrain features to enhance the predictive accuracy of the model. **4) Threshold Determination for Early Warning System (EWS):** The upper and lower susceptibility limits within each classified zone will be analyzed to define critical thresholds. These thresholds will serve as decision-support indicators, forming the foundation of a region-specific Early Warning System (EWS) for potential Glacial Lake Outburst Flood (GLOF) events.

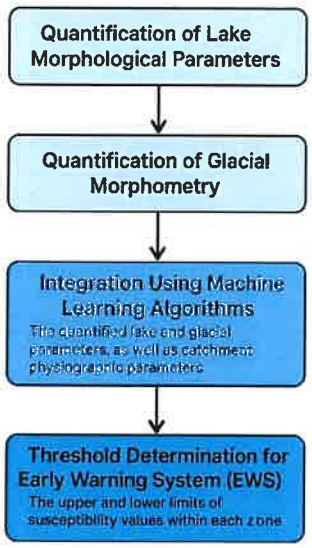


Fig. 7 Proposed Framework for GLOF Susceptibility Mapping and Early Warning System (EWS)

iii. **Validation and Refinement**
- Compare AI/ML model results with historical GLOF events to refine accuracy.
- Utilize expert feedback and field validation for continual improvement.
- Integrate ensemble modeling approaches to enhance reliability.

- **Objective 4: Implementation of a 4G/5G-Based Early Warning System**

i. **Communication Infrastructure**
- Establish a real-time data transmission network utilizing 4G/5G technology.
- Implement redundancy measures such as satellite communication in remote areas.

ii. **Emergency Alert System**
- Develop a cloud-based platform for automated detection of glacial threats.
- Create a multi-channel alert system (SMS, mobile app notifications, sirens, community radio broadcasts).
- Ensure integration with regional disaster management agencies for coordinated response.

iii. **Community Engagement and Training**
- Conduct awareness programs for local communities regarding GLOF risks and response protocols.
- Train field personnel in the use of monitoring tools and emergency communication.
- Develop a user-friendly mobile application for real-time access to alerts and risk assessments.

11. Details of separate new account for the project:

FUND TRANSFER THROUGH PUBLIC FINANCIAL MANAGEMENT SYSTEM (PFMS)
DETAILS OF ORGANIZATION

1.	PFMS Unique Code	SRMUNIV
2 (i)	Agency Name (Name of the University/ Institute/ College, etc)	SRM Institute of Science and Technology
2 (ii)	Agency Type (Statutory Bodies/Autonomous/NGO/Society etc.)	Deemed to be University u/s 3 of UGC Act,1956
2 (iii)	Hierarchy of Agency (Central/State/ District/Block/Tahsil/Panchayat/Village)	Central
2 (iv)	Act/registration No.	F.9-9/98-U.3, U-0473
2 (v)	Date of Registration	02.08.2002
2 (vi)	Registering Authority	MHRD, Govt of India
2(vii)	State of Registration	Tamilnadu
3.	TIN No. (If available)	
4.	TAN No.	CHES17311A
5.	**Complete Contact Address of the Agency**	
5 (i)	Block No./Building/Name of Premises	SRM Nagar
5 (ii)	Road/Street/Post Office	Potheri
5 (iii)	Area/Locality	Kattankulathur
5 (iv)	City/ District	Chengalpattu
5 (v)	State	Tamil Nadu
5 (vi)	Pin code	603203
6.	Contact Person	Dr. S. Ponnusamy
6 (i)	Designation	Registrar
6 (ii)	Phone Number (Land Line)	044-27454646
6 (iii)	Alternate Phone No./Mobile No.	9940036008
6 (iv)	Official E-mail address	registrar@srmist.edu.in
7.	**Bank Account details**	
7(i)	Institution's Account Name (As per bank record)	SRMIST-RESEARCH FUND (SB)
7 (ii)	Account No.	6355923421
7(iii)	IFSC Code	IDIB000G181
7(iv)	Swift Code	IDIBINBBSRP
7(v)	Bank name (in full)	Indian Bank
7(vi)	Branch Name	Indian Bank, SRM University
7(vii)	Complete Branch address	Indian Bank, SRM University, Kattankulathur-603 203.
7(viii)	MICR No.	600019171
7(ix)	Account Type	Savings Account

Certified that the Institute's account is an RTGS/ECS/NEFT enabled branch. I hereby declare that the particulars given above are correct and complete. The above Agency's Account No. and bank details are registered/ mapped under PFMS.

Date: 10/06/24

5/6/24
Registrar
SRM Institute of Science and Technology
SRM Nagar, Kattankulathur - 603 203,
Chengalpattu Dist, Tamil Nadu, India.

Signature of the Competent Authority of the Agency with seal.

Certified that the particulars furnished above are correct as per our records.

Date: 10/06/24.

FOR INDIAN BANK
Manager
SRM University Branch

Signature of the Authorized Bank Official with seal.

12. References:

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13. Date & Month: 26 - June - 2025

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Manoj Kr. Rana.

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15. Signature of Registrar

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Chengalpattu Dist, Tamil Nadu, India.

Signature of Registrar (SRMIST)
Prof. S. Ponnusamy
Registrar
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